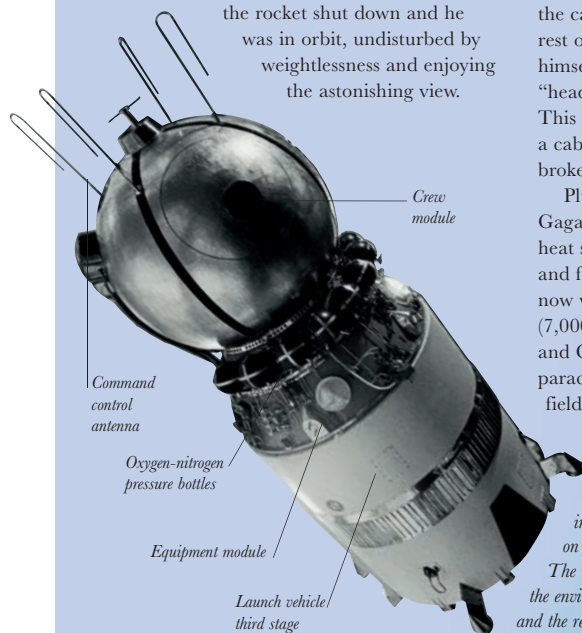


FIRST MANNED SPACE FLIGHT

PRESENTED TO THE WORLD as a smooth and trouble-free triumph of Soviet technology, the first manned space flight was in fact made using equipment that came very close to disaster.

On the morning of April 12, 1961, cosmonaut Yuri Gagarin climbed into the cramped, bare capsule of the Vostok spacecraft on top of its towering launcher. He was left there for 50 minutes, listening to Russian love songs, while preparations for the launch were completed. Then the engines roared and Gagarin felt the g-load build as the vehicle accelerated upward. Minutes later the rocket shut down and he was in orbit, undisturbed by weightlessness and enjoying the astonishing view.



ROCKET POWER

Directly derived from the R-7 which had launched the first Sputnik in 1957, the launcher used to propel Vostok into orbit consisted of a core vehicle and four boosters that dropped off.

DAWN OF AN ERA

The era of manned space flight begins as Vostok 1 blasts off from the Baikonur Cosmodrome on April 12, 1961. Cosmonaut Yuri Gagarin flew for just 1 hour 48 minutes, long enough to circle the globe.

Heading east over the Pacific he plunged precipitately into darkness, crossed South America, and emerged to a new dawn over the Atlantic.

An hour and a quarter into the flight, over Africa, Gagarin heard and felt the retro-rocket ignite to begin reentry. But the capsule failed to separate from the rest of the spacecraft. Gagarin found himself tumbling into the atmosphere, "head, then feet, rotating rapidly." This continued for 10 minutes until a cable burned through and the capsule broke free.

Plunging through the atmosphere, Gagarin heard a cracking noise as the heat shield reached high temperatures and felt the g-load intensify. But all was now well. According to plan, at 23,000ft (7,000m) the capsule hatch blew off and Gagarin's ejector seat fired. He parachuted down into a plowed field near the city of Saratov.

VOSTOK 1

The cosmonaut traveled in the small spherical capsule on top of the Vostok spacecraft. The rest of the craft contained the environmental control system and the retro-rocket which ignited to send the capsule back from orbit.



BRAVE PASSENGER

Gagarin was essentially a passenger in the Vostok spacecraft. Although the controls of the spacecraft were locked, a key had been placed in a sealed envelope in case an emergency situation made it necessary for Gagarin to take control.

"The noise was... like the noise in an aircraft. Then the rocket smoothly, lightly rose..."

YURI GAGARIN
ON HIS FIRST SPACEFLIGHT

